



Cite this: DOI: 10.1039/d2ja00188h

Direct determination of lanthanides by LIBS in REE-rich ores: comparison between univariate and DoE based multivariate calibrations with respect to spectral resolution

Timur F. Akhmetzhanov and Andrey M. Popov *

Laser-induced breakdown spectroscopy (LIBS) was used to detect rare-earth elements (REEs) – Ce, La, Nd, Pr, and Sm – in ores. The problems of significant overlapping of REE lines in LIBS emission spectra and high pairwise correlation between REE contents in certified reference materials (CRMs) of natural origin, decreasing the multivariate model prediction capacity, are addressed in the study. The latter is done by employing a specially designed calibration set (mixtures of REE oxides and $\text{Fe}_2\text{O}_3 + \text{TiO}_2$ as the matrix), based on Latin hypercube sampling providing an extremely low correlation (<0.03) between the contents of different REEs, for the first time for LIBS measurements. Its applicability was assessed for univariate and multivariate (partial least squares regression) calibration models to solve the above problems. The content of each REE in the calibration set varies by two orders of magnitude and completely covers the concentration range in the independent test set of CRMs of natural REE-rich niobium and uranium ores. The selection of analytical lines was carried out through the thermodynamic modeling of LIBS spectra and helped to find the most intensive but not self-absorbed lines in very rich emission spectra of ores. This step greatly reduced the number of required experiments. The comparison of the analytical performance of high- (0.03 nm) and low- (0.26 nm) resolution spectrometers coupled to a high-energy laboratory laser and a high- (0.06 nm) resolution spectrometer coupled to a low-energy portable laser is presented. For the highest resolution spectra, PLS-R models enable the quantitative determination of Ce (0.046–3.95%), La (0.026–2.41%), and Nd (0.015–1.18%) and semi-quantitative determination of Sm (24–1500 ppm). The lowest resolution spectra and PLS-R allow the semi-quantitative determination of La. For the system with a portable laser, PLS-R models enable the determination of Ce (quantitative) and La (semi-quantitative). Pr and Nd were special cases due to considerable matrix effects for atomic lines, addressed by the use of an ionic line of Nd for the highest resolution spectrometer. Reliable Pr ionic lines were not detected due to the strong interference of matrix elements (Fe and Ti). The results show that LIBS has potential to quantify REEs in natural ores.

Received 1st June 2022
Accepted 12th September 2022

DOI: 10.1039/d2ja00188h

rsc.li/jaas

1. Introduction

Rare-earth elements (REEs) have numerous applications in high-tech industries¹ and the determination of REE concentrations is an essential task for geochemical exploration studies, mining, *etc.*² Inductively coupled plasma mass spectrometry (ICP-MS) is a common analytical method used to accomplish this task due to the wide dynamic range and low detection limits.^{3–5} Inductively coupled plasma optical emission spectrometry (ICP-OES) is a feasible alternative to ICP-MS in some cases.^{6,7} However, both techniques require complex sample preparation and equipment, and therefore, cannot be used for rapid analysis outside the laboratory.⁸ Instrumental neutron

activation analysis (INAA) possesses the ability to directly detect trace levels of REEs in geological samples and thus used as a reference to ICP-OES/MS.^{9,10} But time-consuming analysis¹¹ that requires a nuclear reactor is a notable drawback of INAA. Wavelength-dispersive X-ray fluorescence (WDXRF) is neither expensive nor time-consuming and enables REE quantification in solid samples without complex sample preparation and thus used for routine analysis.^{12–17} However, line overlapping and matrix effects (secondary fluorescence and absorption) in WDXRF result in poorer analytical figures of merit in comparison with ICP-OES/MS and INAA.¹⁸ WDXRF spectrometers are also not meant for field analysis. Energy-dispersive XRF (EDXRF) handheld and benchtop spectrometers suitable for field applications are commercially available nowadays. But the sensitivity of EDXRF is lower while line overlapping becomes

Department of Chemistry, Lomonosov Moscow State University, Leninskie Gory, Moscow 119234, Russia. E-mail: popov@laser.chem.msu.ru

much stronger due to lower resolution which complicates obtaining quantitative results.^{19–21}

At the same time, laser-induced breakdown spectroscopy (LIBS) has been proven to be a competitive analytical technique that can enable direct rapid multi-element and most importantly real-time analysis. The use of laboratory LIBS setups allows qualitative and quantitative determination of REEs in solids^{22–24} and liquids.^{25,26} Due to significant spectral interference, the quantification of REEs with LIBS remains a challenging task that was done mostly for samples with relatively simple artificial matrices with the use of univariate^{27–29} and multivariate^{30,31} regression. The latter approach is called “chemometric” and is widely used to find a link between an analyte signal and its concentration hidden behind the interference of different nature.³² In geological materials, light REEs were detected using LIBS.^{33–36} Bhatt *et al.* reported the quantification of REEs with LIBS coupled to chemometric data treatment in geological samples doped with Ce, La and Nd in relatively large amounts (up to 21 wt%).³⁷ The REE spectra are extremely rich and line overlapping can lead to unfortunate misassignment of emission lines known in spectral databases; as a result, inaccurate assignment of spectral lines to REEs can occur as was earlier described by Labutin and Popov.³⁸ Most recently, Rethfeldt *et al.*³⁹ showed the benefits of the chemometric approach to the treatment of LIBS spectra in the quantification of REEs in field geological samples (not doped with REEs). The number of latent variables used to build interval projection on latent structure regression (IPLS-R) models (up to 18 in some cases) seems questionable, though. The reviews^{40,41} regarding small sized field-portable LIBS instruments share the opinion that the transfer of methods developed in the laboratory to field analysis is a necessity. Manard *et al.*⁴² used a SciAps (USA) Z500 handheld LIBS analyzer for a rapid identification of elemental constituents at the sub-percent level in a uranium oxide matrix. Linear calibration curves were plotted for Nd, Yb and Eu. LOD values on the order of hundredths of a percent were determined. The authors concluded that the results can be improved through the modification of the sample preparation.

Aside from strong line overlapping in line-rich REE emission spectra, there also is a problem of strong natural lanthanide correlation in the typical calibration set, consisting of certified standards. The design of experiment (DoE) concept addresses this problem.⁴³ Full factorial design⁴⁴ requires an excessive number of samples (20^5 for the present study) to cover all possible relations of concentrations of analytes to completely fill the hyperspace. Fractional factorial design is able to reduce the number of required experiments.⁴⁵ This can be further developed by exclusion of all correlations between lanthanide concentrations in an orthogonal matrix design proposed by Brereton.⁴⁶ This DoE along with PLS-R was used by Shulyumova *et al.*⁴⁷ to solve the problem of line overlapping (*e.g.*, As/Pb characteristic lines) common for TXRF. This design is, however, best suited for a small concentration range (0–2 mg L^{−1} in ref. 47) due to the number of samples growing as n^2 regardless of the number of factors (*e.g.*, 20^2 in the present work

and 5^2 in ref. 47). This problem was addressed by Kirsanov *et al.*⁴⁸ by the development of a DoE based on the uniform distribution of experimental points in a multidimensional space. The authors have applied this DoE (30 levels and 6 factors), to analyze lanthanides in aqueous solutions using TXRF.⁴⁹ Pairwise correlation remained high (up to 0.2), though. Latin hypercube sampling (LHS), first proposed by McKay *et al.*,⁵⁰ is superior to all abovementioned DoEs as it can be constructed with low correlation for a large number of factors⁵¹ and a tunable number of levels. In our recent study,⁵² the DoE based on LHS (20 levels and 5 factors) with an extremely low correlation (<0.03) enabled the construction of robust PLS-R and principal component regression (PCR) calibration models using artificial mixtures of REE compounds (Fe and Ti oxides as a matrix) that outperformed univariate models providing up to several times better accuracy of Ce, La, Nd, Pr and Sm determination in REE-rich ores and ferromanganese nodules by TXRF and WDXRF.

The goal of this study is to overcome the problems of REE quantification by LIBS. To the best of our knowledge, this is the first study to demonstrate the application of LIBS coupled to chemometrics and a specialized DoE in the quantification of REEs in natural ore samples. In order to find possible ways to simplify the laboratory LIBS setup, we have used two diffraction gratings (2400 and 300 mm^{−1}) mounted onto the turret of one spectrometer, thus comparing the accuracy achieved with high- and low-resolution sets of spectra. For the same reason, the performance of portable diode pumped solid-state and laboratory Nd:YAG lasers was compared. In search of the most suitable methodology, univariate and PLS regressions were compared. We believe that the results of this study will allow cheaper, sensitive, accurate, and direct determination of Ce, La, Nd, Pr, and Sm in ore analysis using LIBS, which can be highly beneficial in the development of instruments for rapid in-line monitoring and geological survey.

2. Materials and methods

2.1. Ores

The first problem this paper deals with is a strong REE correlation (Fig. 1) in the certified reference materials (CRMs) of carbonatite Supergene REE-Nb ore (OREAS 460–465) and U ore (OREAS 100a, 101a, 101b, and 102a) produced by Ore Research & Exploration Pty Ltd (Australia). As the materials constituting OREAS were sourced from a quarry in Western Australia and this correlation is natural for ores. If these samples were used as a calibration set in PLS regression, any deviation from the natural ratio would lie far away from those the model has any information about and therefore can predict.⁴⁶ Thus, we used OREAS samples as a test set. The content of REEs in the set of CRMs is given in Table 1. Note that the certified content of titanium and iron in CRMs was 0.2–6.3 wt% and 4.6–37 wt%, respectively. All the standard samples (in the form of powder) were pressed into 10 pellets ($D = 15$ mm) using a hydraulic press ($P = 5$ MPa). The initial sample preparation used by the company to produce the powders from the natural minerals can be found elsewhere.⁵³

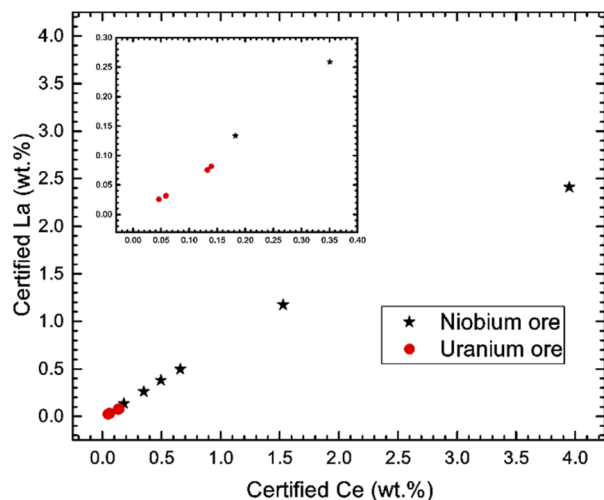


Fig. 1 Correlation between the contents of La and Ce in OREAS certified reference materials.

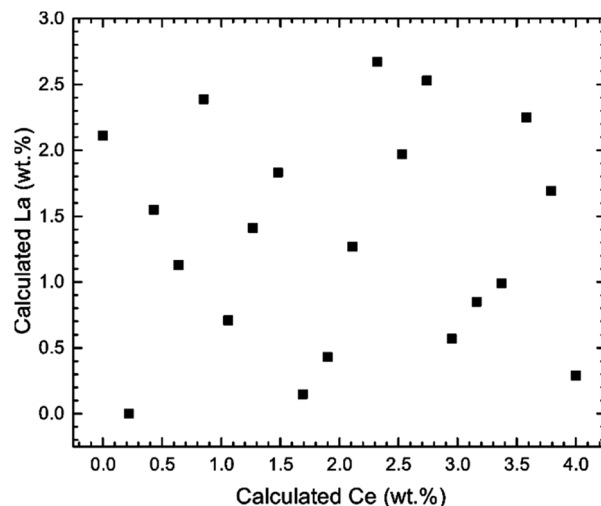


Fig. 2 Correlation between the contents of La and Ce in an artificial set of samples according our DoE based on Latin hypercube sampling.

2.2. Design of experiment

To solve the problem of REE correlation in natural samples of ores, we used a DoE based on Latin hypercube sampling, thoroughly described elsewhere.⁵⁰ This is a multilevel multi-factor design allowing a virtually orthogonal matrix. Only such a matrix (see the difference between Fig. 1 and 2) can be used for the simultaneous determination of REEs at any their ratios following recommendations given by Brereton.⁵⁴ In this work we used 20 levels of concentrations (samples) for 5 factors (Ce, La, Nd, Pr, and Sm) to obtain a very low pairwise correlation coefficient (<0.03). The bottleneck here is the wide concentration range (2 orders of magnitude) that we wanted to cover. Other than that, an even smaller number of samples are required for 5 factors with a slightly higher correlation (e.g., 0.05 for 10 levels and 5 factors). In addition, our correlation limit (0.03) was almost an order of magnitude better than those for the DoE proposed by Kirsanov *et al.*⁴⁸ The content range of every analyte in our DoE (see Table 2) completely covers those for ores (see Table 1) and seems to be suitable for analytical practice. Note that this paper considers only REE-rich ores and therefore

relatively high concentrations are used in the DoE (up to 4%; depending on the element, the same way as the real ores).

2.3. Calibration set

To produce a set of samples corresponding to our DoE, powders of CeO_2 (99.95%), La_2O_3 (99.99%), Nd_2O_3 (99.99%), Pr_6O_{11} (99.9%) and Sm_2O_3 (99.9%) purchased from Sigma Aldrich (Germany) were carefully mixed in different ratios. Oxide powders of Fe_2O_3 (chemical grade) and TiO_2 (chemical grade) produced by Rechem (Russia) roughly simulated the matrix of a test set with concentrations of 62–67 wt% and 1.8 wt% in the mixture, respectively. The contents of REEs in the calibration set are given in Table 2. All samples of the calibration set (in the form of powder) were pressed into 20 pellets ($D = 15$ mm) using a hydraulic press ($P = 5$ MPa).

2.4. Equipment

Two LIBS setups were used for the comparison of the analytical performance of univariate and PLS regression for REE

Table 1 Concentrations of REEs (in mg kg^{-1}) in an independent test set of certified samples of OREAS ores

Standards	Ce	La	Nd	Pr	Sm
Niobium ores					
OREAS 460	1830 $\pm$ 10	1330 $\pm$ 10	800 $\pm$ 30	240 $\pm$ 10	104 $\pm$ 5
OREAS 461	3510 $\pm$ 10	2590 $\pm$ 10	1630 $\pm$ 40	480 $\pm$ 30	214 $\pm$ 8
OREAS 462	4950 $\pm$ 100	3790 $\pm$ 30	2560 $\pm$ 70	740 $\pm$ 20	369 $\pm$ 5
OREAS 463	6590 $\pm$ 70	4966 $\pm$ 70	3610 $\pm$ 70	1000 $\pm$ 20	538 $\pm$ 4
OREAS 464	15 300 $\pm$ 300	11 700 $\pm$ 100	9940 $\pm$ 160	2600 $\pm$ 60	1500 $\pm$ 20
OREAS 465	39 500 $\pm$ 800	24 100 $\pm$ 500	11 800 $\pm$ 300	3770 $\pm$ 100	1360 $\pm$ 20
Uranium ores					
OREAS 100a	460 $\pm$ 20	260 $\pm$ 10	152 $\pm$ 8	47 $\pm$ 2	24 $\pm$ 0.4
OREAS 101a	1400 $\pm$ 90	820 $\pm$ 30	400 $\pm$ 20	134 $\pm$ 7	49 $\pm$ 1
OREAS 101b	1330 $\pm$ 70	750 $\pm$ 30	390 $\pm$ 20	127 $\pm$ 6	48 $\pm$ 3
OREAS 102a	590 $\pm$ 30	320 $\pm$ 10	180 $\pm$ 9	58 $\pm$ 3	25 $\pm$ 0.7

Table 2 Concentrations (in wt%) of REEs in an artificial set of calibration samples

No. mixture	Ce	La	Nd	Pr	Sm
1	3.37	0.99	0.07	0.27	0.23
2	1.06	0.71	0.35	0.32	0.17
3	0	2.11	1.19	0.21	0.34
4	1.27	1.41	0.42	0.25	0.04
5	0.22	0	0.63	0.09	0.38
6	2.95	0.57	1.05	0.30	0.25
7	2.32	2.67	0.98	0.11	0.21
8	2.74	2.53	0.49	0.17	0.06
9	1.48	1.83	0.91	0.13	0.02
10	3.58	2.25	0.56	0.15	0.36
11	2.53	1.97	0.28	0.38	0.27
12	3.16	0.85	1.26	0.04	0
13	0.43	1.55	0.14	0	0.13
14	0.85	2.39	0.70	0.40	0.11
15	4.00	0.29	1.12	0.34	0.09
16	2.11	1.27	0.21	0.23	0.30
17	1.90	0.43	0	0.19	0.19
18	3.79	1.69	0.84	0.02	0.40
19	1.69	0.15	0.77	0.06	0.15
20	0.64	1.13	1.33	0.36	0.32

determination in ores. For the clarity of reading, we label these setups with numbers 1–3 (the first setup has two diffraction gratings as described below) in the order presented in Table 3. The first experimental setup is a conventional laboratory LIBS setup. It is due to the use of a lamp-pumped, water-cooled laser and bulky spectrometer, both components are relatively expensive and fragile. We used the second harmonic of a LOTIS “LS-2134 UTF” (LOTIS TII, Belarus) water-cooled flash-lamp pumped Q-switched Nd:YAG laser (pulse duration ~ 6 ns and beam diameter ~ 6 mm). The laser beam was directed using a system of mirrors ($d = 25.4$ mm) coated in the range of 524–532 nm and quartz prisms. The beam was focused perpendicularly to the sample surface with a coated (532/1064 nm) lens ($f = 250$ mm and $d = 50.8$ mm). Plasma emission was focused onto the entrance slit (25 μm) of a Czerny–Turner spectrograph MS5204i from SOL instruments with a spectrograph with diffraction gratings 300 and 2400 mm^{-1} using a two-lens condenser perpendicular to the laser beam. An intensified CCD camera “NANOGATE-2v” (Nanoscan, Russia, described elsewhere⁵⁵) recorded the spectra.

The second experimental setup can be characterized as a cheaper option that can be implemented as a desktop system similar in size and capabilities to energy dispersive XRF

spectrometers. We used a Laser-Compact “TECH-527 Advanced” (Laser-Compact, Russia) portable air-cooled Nd:YVO₄ DPSS laser ($\lambda = 527$ nm and pulse duration ~ 4 ns). A beam expander with a fixed factor of 8 was used to expand a relatively narrow laser beam (~ 0.6 mm). The beam passed through a system of dichroic mirrors ($d = 25.4$ mm) and was perpendicularly focused on a sample surface by an N-BK7 plano-convex quartz lens ($f = 200$ mm) into a spot with a diameter ~ 50 μm . The central part of a plasma image was projected by an optical system consisting of two 90° off-axis parabolic mirrors with through holes (Thorlabs, USA), an aspherical lens, and a circle-line optic fiber cable onto a 25 μm slit of a high aperture $f/4$ Czerny–Turner spectrograph “HR320” (ISA Instruments, USA).

2.5. Selection of analytical lines

Careful selection of analytical lines for the analysis of ores containing Fe, Ti, and REEs each having a very rich spectrum is a challenging task. We used thermodynamic modeling of the emission spectrum to overcome the difficulties. The modeling procedure is thoroughly described elsewhere.⁵⁶ Here we only state the initial assumptions for the laser-induced plasma spectra modeling as follows:

- (1) Complete atomization of an evaporated part of a sample.
- (2) Equality of element mass fractions in plasma and the sample.
- (3) Plasma is under local thermodynamic equilibrium conditions, uniform, stationary and electrically neutral.

Since we wanted to find the most intensive lines free of self-absorption, the idea was to use the spectrum of early stages ($T = 0.9$ eV and $N_e = 8 \times 10^{16} \text{ cm}^{-3}$) of plasma evolution to find appropriate ionic lines. This idea seems even more appropriate given the fact that the ionization potentials of REEs ($E_{\text{ion}}(\text{La}) = 5.61$ eV) are almost as low as those of alkaline metals ($E_{\text{ion}}(\text{Na}) = 5.14$ eV). Naturally, ionic lines are expected to be dominant in the spectrum at the early stages of plasma evolution. A comparison of experimental and model spectra used in this work is presented in Fig. 3. As one can see the model is in good agreement with the experiment and therefore the results are trustworthy. The differences occurring in Fig. 3 are due to spectra acquisition parameters

Table 3 Main differences in experimental LIBS setups used in the work

No	Lasers		Spectrometers	
	E , mJ	Rep. rate, Hz	Spectral range, nm	Resolution, nm
1	26	5	360–500	0.03
2	0.4	200	430–500	0.06
3	26	5	360–500	0.26

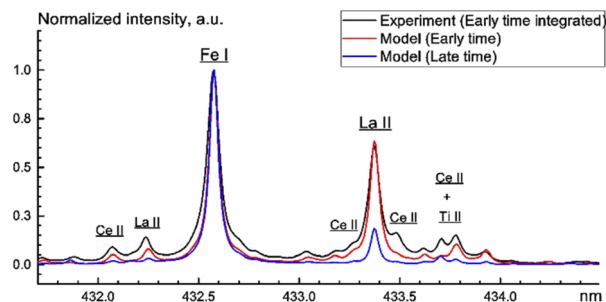


Fig. 3 A comparison of LIBS spectra simulated both for the early stage ($T = 0.9$ eV and $N_e = 8 \times 10^{16} \text{ cm}^{-3}$) of plasma evolution (red) and the late stage ($T = 0.6$ eV and $N_e = 4 \times 10^{16} \text{ cm}^{-3}$) (blue) with an experimental spectrum (black) obtained for setup 1. All spectra are for the OREAS 465 sample.

that we used to reach the maximum signal-to-noise ratio. The experimental parameters used in this work are presented in Table 4 along with analytical lines and their parameters. For every line chosen, a parameter evaluating possible self-absorption was less than 10% for the highest content of every lanthanoid. Notice that there are two cases when we used an atomic line instead of ionic. Because of a relatively low content of Pr, signal-to-noise ratio for all Pr lines was less than 40, and we were not able to find enough intense lines of Pr. In the spectra of a short-live DPSS laser induced plasma of ores, Nd ionic lines were either very weak to be detected or completely overlapped by more intense Fe I, II and Ti I, II lines despite Nd having a three times higher content than Pr. Therefore, we used Nd atomic line at 494.48 nm which was predicted by the simulation as the most intense line of neodymium under our conditions. For this reason, we used a relatively longer gate and delay (Table 4). For setup No. 2, the delays are shorter (up to an order of magnitude) due to the lower pulse energy of the DPSS laser leading to a short-lived plasma as we showed in a previous study where the same laser was used.⁵⁷ Earlier delay leads to notably higher line widths causing additional line overlapping in LIBS emission spectra which is another reason to use the chemometric approach to data treatment.

2.6. Multivariate regression

We aimed to solve the problem of not just lanthanide correlation but also the problem of line overlapping. For this reason, we used a multivariate regression method called Projection on Latent Structures (PLS-R). As it is a well-established and described^{54,58,59} method, we only briefly explain its idea and more thoroughly the way we used it to solve the problem of line overlapping. This algorithm applies the idea of dimensionality reduction combined with the exclusion of all irrelevant information from a given number of wavelengths through the search for the highest correlation between the information stored in the spectral data and data on the concentration of an element of interest. Thus, an information about that how the intensities at various wavelengths are correlated with concentrations of REEs can be reduced from tens or hundreds of values within a given spectral window (at many wavelengths) to two-three principal components. And only those structures were used to build a regression model. To find the right number of latent structures, we used the Root-Mean Square Error of Cross-Validation (RMSECV) along with the information we have from the spectral

modeling algorithm that can tell us how many elements can give strong emission signals in the chosen range of wavelengths. Since we have the DoE that must not be dramatically changed (by exclusion of many samples at once) and the number of calibration samples is relatively small, leave one out (LOO) cross-validation was used. We considered that the right number of components is the one that characterizes only the link between intensity and concentration.

PLS-R does not have any variable parameters in it, *e.g.*, the shape of the contour (as for deconvolution – Gauss, Lorentz, *etc.*) or the positions of analytical and interfering lines. For this reason, a wide spectral range containing many fluctuating signals can lead to correlations between the intensities of irrelevant elements and concentrations of relevant elements in the calibration set that will not be reproduced in the test set. That would be the case of negative influence of false correlations on the multivariate regression model making it “unstable”. To make the multivariate model more robust, we must very carefully exclude all irrelevant information. In order to do so, we used the variable (wavelength) selection technique.^{60,61} It means that one must only narrow spectral intervals in PLS-R manually cutting off lines of other elements (especially off-DoE elements). And this leaves only the intervals that contain the analytical line overlapped with the interfering ones (Table 5). Another important step in data pretreatment is background (baseline) subtraction.⁶² Emission signals used in this work can be characterized as very noisy in terms of a shifting and tilting baseline mostly coming from the wings of lines of matrix elements (Fe and Ti) and plasma continuous spectra natural for early delays. The main difficulty here is to preserve the information on the concentration of the analyte and for this reason we excluded the baseline in a way that in calibration and test samples the resulting spectra closely resembled each other in the baseline level. It is easy to see that the baseline is mostly the problem for the elements in the low concentration range (Pr 50–150 ppm and Sm 20–100 ppm). Thereafter all models were tested on the spectra of OREAS samples (test set). Predicted–measured plots that can give general understanding of the quality of the models. We did not use scaling for emission spectra to avoid increasing noise and worsening of multivariate models. We compared the analytical performance of PLS-R to that of univariate regression based on peak intensities (after background exclusion) of analytical lines. This way of data treatment is simple and commonly used in analytical practice.

Table 4 Parameters of analytical lines used in this work and experimental conditions of spectra acquisition

Experimental setup	Line, nm	E_{low} , eV	E_{up} , eV	g_{λ} , 10^8 s^{-1}	Delay, ns	Gate, ns
No 1	Ce II 462.81	0.516	3.195	4.25	500	1500
No 2					150	150
No 1	La II 433.38	0.173	3.033	3.1	1000	1000
No 2					50	1000
No 1	Nd II 406.11	0.139	2.647	8.7	1000	1000
No 2	Nd I 494.48	0.471	3.524	14	600	1000
No 1	Pr I 495.14	0	2.503	8.4	7500	5000
No 1	Sm II 359.26	0.378	3.829	9.2	2000	2000

Table 5 Spectral ranges used in this work and known interferences

Experimental setup	Line, nm	Interfering lines identified in model spectra, nm	Spectral ranges, nm
No 1	Ce II 462.81	Ti I – 462.82, 462.93, Ti II – 462.93, Fe I – 462.93	462.67–462.88
No 2			462.62–463.02
No 1	La II 433.38	Fe I – 432.58, Ce II – 433.18, 433.27, 433.62	433.10–433.60
No 2			433.19–433.61
No 3			432.99–434.47
No 1	Nd II 406.11	Ti I – 406.03, Fe I – 406.36, Ce II – 406.22	406.02–406.16
No 2	Nd I 494.48	Ce II – 494.46, La I – 494.58, Fe I – 494.64, 495.73, 495.76, La II – 494.65	492.25–494.54
No 1	Pr I 495.14	La I – 494.98, 495.21, Ce I – 495.19, Fe I – 495.73, 495.76	495.05–495.19
No 1	Sm II 359.26	Fe I – 359.27, 359.46, Nd II – 359.26, Ti II – 359.31, Cr I – 359.33 (in ores)	359.09–359.26

MATLAB (MathWorks Inc.) software was used for data pre-treatment and univariate regression. PLS-R models were obtained by using The Unscrambler X (Version 10.4, CAMO Software). Graphics were plotted by using the OriginPro® software package.

3. Results and discussion

3.1. Emission spectra

The complicated structure of LIBS emission spectra is clearly presented in Fig. 4 and shows a good example of line

overlapping. To minimize pulse-to-pulse fluctuation, for all three experimental setups no less than 50 acquisitions were made and after the exclusion of outliers (by the interquartile range) they were averaged to get a single spectrum. As one can see, the most intensive line belongs to Fe and its proximity to La can have a negative impact on the signal to noise ratio in terms of chemometrics, mostly by drowning the signal of lanthanide in its wing. This negative influence cannot be reduced by variable selection or baseline subtraction (but only by higher spectral resolution which has its own limitations), and thus in this work we considered the line as analytical only in the case of weak overlapping with elements not included in the DoE (see Fig. 4). Generally, this overlapping cannot be completely avoided and that is the reason we used PLS-R that searches for the correlation between signal intensity and concentration and not only the direction of the most variance produced by the signal intensity. In this work we aimed to use lines that are more intensive than the possible interference found in the model spectra. It gives us a chance to cover more variance of signal intensity in the narrow regions we used (Table 5). This way of line selection allowed us to obtain relatively simple multivariate regression models composed of 2–3 latent structures (principal components or PCs) depending only on the experimental data. This lies in good agreement with a similar case described in the literature,⁶³ where the authors used a narrow spectral region (405.62–405.96 nm of 26 pixels) and thus obtained robust multivariate models composed of 2–3 PCs.

Low-resolution LIBS spectra are the most problematic even for the chemometric approach to data treatment. It includes all the abovementioned problems. For this reason, we used the same highly intensive line La II 433.38 nm to compare the performance of different spectrometers and regression techniques. We also checked the accuracy on Ce II 462.81 nm but were not able to build a reliable PLS-R model due to false correlations with Fe and Ti lines (Table 5). All other lanthanides with lower concentrations have shown even lower potential for any regression technique.

3.2. Univariate and multivariate calibrations in LIBS

Multivariate regression models were built through the combination of two assumptions:

- (1) RMSECV must drop or be in a local minimum.

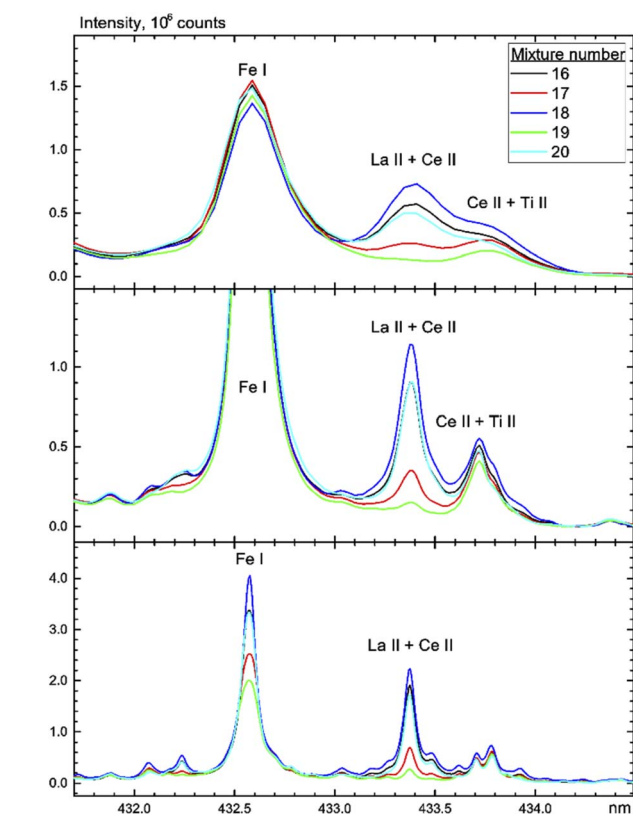


Fig. 4 LIBS spectra of five calibration samples in the 433 nm region for setup 3 (top), setup 2 (middle), and setup 1 (bottom). There are two kinds of spectral interference – lanthanoid lines overlap with each other and with matrix elements (mostly Fe and Ti).

(2) The number of PCs must not exceed the number of elements, lines of which overlap in a selected region.

We believe that these assumptions consider both mathematical explanation of calibration and have physical meaning. The first means that our model can predict itself and the second can guarantee that it only uses the shapes of overlapping lines excluding false correlations with the baseline or matrix elements. And since calibration samples are artificial, they have a predefined number of elements. This procedure was used in all cases to avoid overfitting. For example, note that in the case of Ce II 462.81 nm the number of components used in this work is different for two experimental setups (Table 6). It is due to different spectral resolution. For higher resolution, we were able to reduce the influence of signals of matrix elements through variable selection, which led to a reduced number of PCs in comparison to a setup with lower spectral resolution. We demonstrate prediction plots obtained using univariate and PLS regression techniques in Fig. 5 (top) and Fig. 6 (top). Note that for univariate regressions, the error bars (Δ) of mean values of the analyte concentrations were calculated with the use of relative standard deviation (RSD) of the analyte signal for a two sided t -distribution as follows:

$$\Delta = \frac{t(P, f) \times \text{RSD}}{\sqrt{n}}, \quad (1)$$

where P is the confidence level (0.95), and f and n are the degrees of freedom ($n - 1$) and the number of averaged spectra, respectively. In this way, the Δ and the RMSEC are the estimators of precision and accuracy of the analysis respectively. In both cases we aimed to show Ce as an element determined with the two of our experimental setups. This element has the largest content in calibration and test sets, and also the analytical line was the same in both cases. Thus, we can compare the performance of both analytical systems. As one can expect, PLS-R coupled to a higher resolution spectrometer gives the most accurate results

Table 6 The number of principal components (PCs), the raw root mean square errors of calibration (RMSEC, mg kg⁻¹) and that weighed by the average concentration for the given concentration range (RMSEC/ $\bar{C}$, %) of different calibration strategies: univariate and partial least squares regression

Element	Univariate		PLS-R		
	RMSEC	RMSEC/ $\bar{C}$	PCs	RMSEC	RMSEC/ $\bar{C}$
High-powered laser + high resolution spectrometer					
Ce	1542	8	2	888	4
La	1466	11	2	1171	9
Nd	1010	15	3	942	14
Pr	348	17	2	346	17
Sm	336	17	2	256	13
Low-powered laser + high resolution spectrometer					
Ce	2030	10	3	1567	8
La	1038	8	2	994	7
Nd	1234	18	2	982	15
High-powered laser + low resolution spectrometer					
La	1053	8	2	846	6

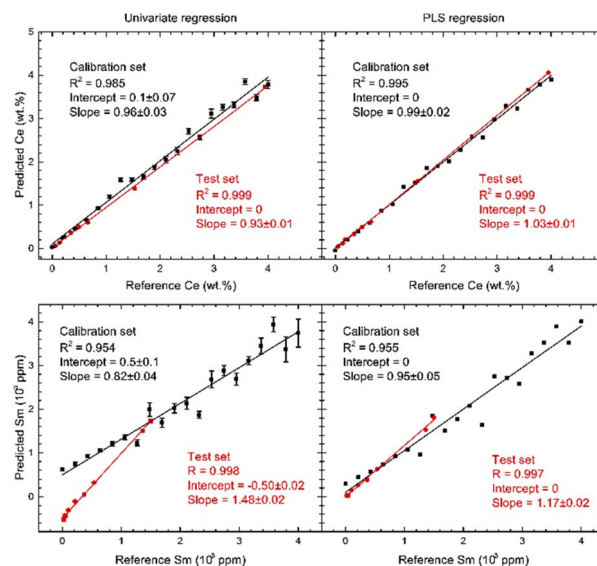


Fig. 5 Ce (top) and Sm (bottom) prediction plots (setup 1) for univariate (left) and PLS (right) regression. Black squares and red circles correspond to calibration and test sets, respectively. Black and red lines mark the best linear fits of the calibration and test results, respectively. Fit parameters are shown in a legend.

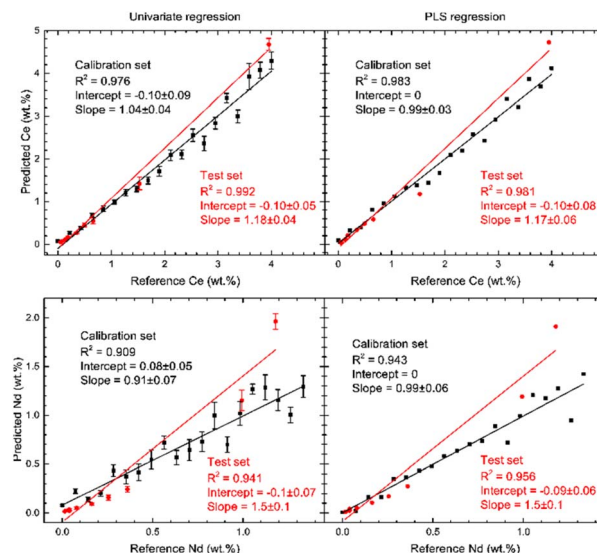


Fig. 6 Ce (top) and Nd (bottom) prediction plots (setup 2) for univariate (left) and PLS (right) regression. Black squares and red circles correspond to calibration and test sets, respectively. Black and red lines mark the best linear fits of the calibration and test results, respectively. Fit parameters are shown in a legend.

in terms of RMSEC (Table 6). And the lower resolution with univariate regression gives the poorest results.

Notably the RMSEC of the lower resolution system coupled to chemometrics is very close to the one obtained with the higher resolution spectrometer with univariate regression. It clearly shows the limitations one can encounter using a higher resolution spectrometer to solve the problem of line overlapping. In this case, one can see that the synergy of the

equipment and data treatment allows the reduction of RMSEC by 2.3 times. Another case of interest regards poorer calibration plots presented in Fig. 5 (bottom) and Fig. 6 (bottom). Note that in both cases, the slope of the calibration curve demonstrates that the univariate model gives poorer results. At the same time, PLS-R models are reliable even for the worst cases. The parameters of calibrations are given in Table 6. As one can see in all cases LIBS coupled to chemometrics shows either comparable or up to two times better results in terms of less RMSEC. To address the difference in concentration ranges for different REEs, we present $\text{RMSEC}/\bar{C}$. This value is two-three times lower for Ce as it has high concentrations (up to 4 wt%) and La due to the lower $Z(T)$ value (partition function), which in both cases results in high intensity of the emission lines. It should be noted that PLS-R can only assist in building a robust regression model but it cannot compensate for matrix effects nor can it accurately extract the signal of an analyte from the contour of the more intensive line of matrix elements (Fe, Ti, *etc.*). The latter can only be done with the use of a higher resolution spectrometer. Therefore, we stress that the possibility of REE determination depends on spectral resolution yet the accuracy (in the absence of strong matrix effects) depends on the regression technique. Thus, we can conclude that PLS-R is a more reliable way to build a robust calibration model in LIBS that can in some cases improve the accuracy.

3.3. A comparison of the prediction capacity of univariate and multivariate regression models

We present a comparison of the concentrations of all determined REEs in our independent test set of OREAS ore samples to certified concentrations in Table 1. In order to show the advantages of the chemometric approach we calculated the relative error as follows:

$$\text{Relative error} = 100 \times (C_{\text{predicted}}/C_{\text{certified}} - 1), \quad (2)$$

where $C_{\text{predicted}}$ and $C_{\text{certified}}$ are the analyte content predicted by the calibration model discussed above and the certified content

(Table 1), respectively. This figure of merit gives a clearer view on the performance of both regression techniques than the RMSEP. The latter is more influenced by the errors in a higher concentration range, while most of the ore samples have a relatively low REE content (Table 1) unlike the samples from the calibration set (uniformly distributed within the concentration range). Fig. 7 shows that despite the presence of outliers (which can be associated with matrix effects) the overall performance of PLS-R in the prediction of REE concentration in niobium and uranium ores is better compared to univariate regression. The most notable fact is the ability of PLS regression to keep the accuracy in the range of about $\pm 10\%$ (Ce, La, and Nd) for the experimental setup composed of a high-powered flash-lamp pumped Nd:YAG laser and high-resolution spectrometer (setup 1). The results for Pr and Sm for this setup are mostly semi-quantitative (uncertainty range of about $\pm 30\%$) and show considerable scattering (note that the concentration range for both elements is one order of magnitude less than Ce but scattering does not grow as fast) and still PLS-R shows better accuracy than the univariate regression.

Significant accuracy improvement is demonstrated for this laser in combination with the low-resolution spectrometer (setup number 3) in La determination which is not possible (even semi-quantitative) without the chemometric approach to data treatment. As REEs are naturally correlated in these samples (Fig. 1), it might be possible to use a cheaper spectrograph with less resolving power in the LIBS system to monitor the contents of REEs in niobium and uranium ores. Experimental setup 2 allowed the quantitative determination of Ce and notable accuracy improvement for La. The determination of Nd was less successful and no accuracy improvement has been achieved as can be seen on both the accuracy plot (Fig. 6, bottom) and the box plot (Fig. 7). We believe that this case is similar to Pr determination with setup number 1 as in both cases the atomic line was used as analytical. The use of the Nd ionic line showed improved results for the Nd:YAG laser. The latter seems reasonable since REEs can be easily ionized and as the form of the change of their content is different in calibration and test sets there might be

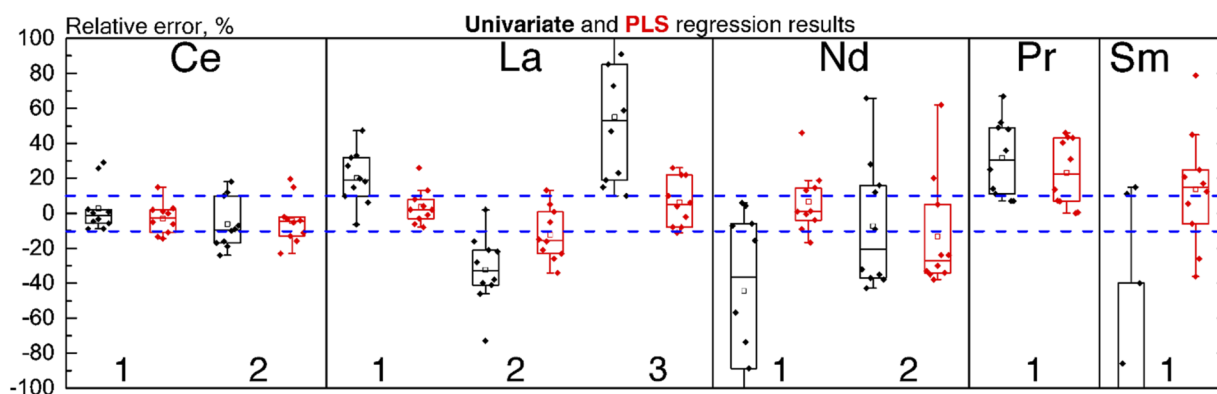


Fig. 7 Relative errors of different calibration strategies and experimental setups for ore CRMs analysed with: univariate (black boxes) and PLS-R (red boxes) regression. Dashed blue line corresponds to the deviation of $\pm 10\%$ as a range of quantitative analysis. Numbers 1–3 correspond to the numbers of experimental setups (see Table 3). Boxes show the interquartile range (IQR), whiskers indicate the highest/lowest values between $Q1 - 1.5IQR$ and $Q3 + 1.5IQR$, and small squares in boxes and solid lines indicate average and median values respectively.

considerable differences between signal intensities in the sets at the late stages of plasma evolution (matrix effects). Hence, we can conclude that ionic lines are a more reliable source of spectral data for REE determination in ores and characterizes with less systematic error. A different experimental study is required to give more detailed explanation of this interesting finding, which lies beyond the topic of this paper since only spectral resolution (along with the optimization of some experimental parameters described in Sections 2.4, 2.5) and regression techniques are investigated here.

4. Conclusions

We used LIBS for ore direct analysis for REE determination. We aimed to solve two problems of REE quantification that reinforce each other: (1) the negative influence of strong lanthanide correlation in natural OREAS ore samples and (2) significant spectral interference. We showed that artificial samples made of iron and titanium oxides as a matrix and 5 REE oxides (La, Ce, Pr, Nd, and Sm) mixed in accordance with the design of experiment based on Latin hypercube sampling (20 levels and 5 factors) with an extremely low correlation (<0.03) allowed us to obtain robust multivariate calibration models (RMSEC for PLS-R in all cases was comparable to univariate models or lower up to 2 times). It was shown that only higher spectral resolution can provide the possibility of REE determination by LIBS in these samples but the accuracy of determination strongly depends on the regression technique, especially for a low concentration range. It was found that the differences in the relation of the REE content in calibration and test sets can affect the accuracy (significant systematic error) for atomic analytical lines and improvement was achieved with the use of an ionic line (Nd case). We demonstrated that PLS regression provided better analytical performance for La, Nd and Sm quantification in ores rather than univariate regression. In addition, it was shown that even a low-resolution spectrometer can be a powerful tool in La determination when LIBS is coupled with chemometrics. We believe that it can allow the simplification of the experimental setup in the case of samples with natural lanthanide correlation where the content of La can be used for inline control and decision making. For the same purpose, a cheaper DPSS laser can be used because it demonstrates accuracy in the range of quantitative analysis for Ce and semi-quantitative for La but only with the use of PLS regression. The quantitative determination of the concentrations of minor REEs (Pr and Sm) using LIBS still remains an issue to be solved in the future. In addition, a more detailed study on the influence of line selection (atomic/ionic) on the accuracy is essentially required.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

The authors thank the Russian Foundation for Basic Researches (grant no. 19-33-90242) for financial support. The authors also

thank Dr Timur A. Labutin and Mr Ivan N. Krylov for fruitful discussions on artificial sample set design and preparation.

References

- 1 V. Balaram, *Geosci. Front.*, 2019, **10**, 1285–1303.
- 2 J. R. Bacon, O. T. Butler, W. R. L. Cairns, J. M. Cook, R. Mertz-Kraus and J. F. Tyson, *J. Anal. At. Spectrom.*, 2019, **34**, 9–58.
- 3 B. Zawisza, K. Pytlakowska, B. Feist, M. Polowniak, A. Kita and R. Sitko, *J. Anal. At. Spectrom.*, 2011, **26**, 2373–2390.
- 4 Q. Ma, M. Yang, H. Zhao, N. J. Evans, Z. Y. Chu, L. W. Xie, C. Huang, Z. D. Zhao and Y. H. Yang, *J. Anal. At. Spectrom.*, 2019, **34**, 1256–1262.
- 5 X. Liao, Z. Hu, T. Luo, W. Zhang, Y. Liu, K. Zong, L. Zhou and J. Zhang, *J. Anal. At. Spectrom.*, 2019, **34**, 1126–1134.
- 6 F. O. Leme, L. C. Lima, R. Papai, N. Akiba, B. L. Batista and I. Gaubeur, *J. Anal. At. Spectrom.*, 2018, **33**, 2000–2007.
- 7 B. Ticová, K. Novotný and V. Kanický, *Chem. Pap.*, 2019, **73**, 2913–2921.
- 8 S. K. Pradhan and B. Ambade, *J. Anal. At. Spectrom.*, 2020, **35**, 1395–1404.
- 9 H.-G. Stosch, *Phys. Sci. Rev.*, 2016, **1**, 1–25.
- 10 K. F. R. Damascena, R. dos S. Amaral, J. A. dos Santos, F. A. Genezini, A. A. da Silva and R. S. C. Menezes, *J. Radioanal. Nucl. Chem.*, 2015, **304**, 1053–1058.
- 11 A. P. Ribeiro, A. M. G. Figueiredo and J. B. Sígolo, *J. Radioanal. Nucl. Chem.*, 2005, **263**, 645–651.
- 12 R. Sitko, B. Zawisza and M. Czaja, *J. Anal. At. Spectrom.*, 2005, **20**, 741–745.
- 13 D. Suvorova, E. Khudonogova and A. Revenko, *X-Ray Spectrom.*, 2017, **46**, 200–208.
- 14 W. Wu, T. Xu, Q. Hao, Q. Wang, S. Zhang and C. Zhao, *J. Rare Earths*, 2010, **28**, 30–36.
- 15 K. Nakayama and T. Nakamura, *Anal. Sci.*, 2005, **21**, 815–822.
- 16 E. De Pauw, P. Tack, M. Lindner, A. Ashauer, J. Garrevoet, B. Vekemans, G. Falkenberg, F. E. Brenker and L. Vincze, *Anal. Chem.*, 2020, **92**, 1106–1113.
- 17 I. E. De Vito, R. A. Olsina and A. N. Masi, *Fresenius. J. Anal. Chem.*, 2000, **368**, 392–396.
- 18 A. Smoliński, M. Stempin and N. Howaniec, *Spectrochim. Acta Part B At. Spectrosc.*, 2016, **116**, 63–74.
- 19 B. Lemièrre, *J. Geochem. Explor.*, 2018, **188**, 350–363.
- 20 C. Vanhoof, J. R. Bacon, A. T. Ellis, L. Vincze and P. Wobrauschek, *J. Anal. At. Spectrom.*, 2018, **33**, 1413–1431.
- 21 G. E. M. Hall, G. F. Bonham-Carter and A. Buchar, *Geochem. Explor. Environ. Anal.*, 2014, **14**, 99–123.
- 22 V. K. Unnikrishnan, R. Nayak, P. Devangad, M. M. Tamboli, C. Santhosh, G. A. Kumar and D. K. Sardar, *Mater. Lett.*, 2013, **107**, 322–324.
- 23 K. M. Abedin, A. F. M. Y. Haider, M. A. Rony and Z. H. Khan, *Opt Laser. Technol.*, 2011, **43**, 45–49.
- 24 M. Z. Martin, S. Allman, D. J. Brice, R. C. Martin and N. O. Andre, *Spectrochim. Acta Part B At. Spectrosc.*, 2012, **74–75**, 177–183.
- 25 D. Alamelu, A. Sarkar and S. K. Aggarwal, *Talanta*, 2008, **77**, 256–261.

- 26 X. Yang, Z. Hao, M. shen, R. Yi, J. Li, H. Yu, L. Guo, X. Li, X. Zeng and Y. Lu, *Talanta*, 2017, **163**, 127–131.
- 27 K. G. Myhre, M. J. Mehta, M. Z. Martin and M. Du, *Spectrochim. Acta Part B At. Spectrosc.*, 2018, **149**, 30–34.
- 28 D. A. Hartzler, C. R. Bhatt, J. C. Jain and D. L. McIntyre, *J. Energy Resour. Technol. Trans. ASME*, 2019, **141**, 1–14.
- 29 C. R. Bhatt, D. Hartzler, J. C. Jain and D. L. McIntyre, *Opt Laser. Technol.*, 2020, **126**, 106110.
- 30 M. Z. Martin, R. C. Martin, S. Allman, D. J. Brice, A. Wymore and N. Andre, *Spectrochim. Acta Part B At. Spectrosc.*, 2015, **114**, 65–73.
- 31 C. R. Bhatt, F. Y. Yueh and J. P. Singh, *Appl. Opt.*, 2017, **56**, 2280.
- 32 R. G. Brereton, *Chemometrics: Data Analysis for the Laboratory and Chemical Plant*, Wiley, New York, 2003.
- 33 T. A. Labutin, S. M. Zaytsev, A. M. Popov and N. B. Zorov, *J. Anal. At. Spectrom.*, 2016, **31**, 2223–2226.
- 34 C. Fabre, D. Devismes, S. Moncayo, F. Pelascini, F. Trichard, A. Lecomte, B. Bousquet, J. Cauzid and V. Motto-Ros, *J. Anal. At. Spectrom.*, 2018, **33**, 1345–1353.
- 35 S. Romppanen, H. Häkkinen and S. Kaski, *Spectrochim. Acta Part B At. Spectrosc.*, 2017, **134**, 69–74.
- 36 S. Müller, J. A. Meima and D. Rammlmair, *J. Geochem. Explor.*, 2021, **221**, 106697.
- 37 C. R. Bhatt, J. C. Jain, C. L. Goueguel, D. L. McIntyre and J. P. Singh, *Appl. Spectrosc.*, 2018, **72**, 114–121.
- 38 T. A. Labutin and A. M. Popov, *Fuel*, 2016, **167**, 375–376.
- 39 N. Rethfeldt, P. Brinkmann, D. Riebe, T. Beitz, N. Köllner, U. Altenberger and H.-G. Löhmansröben, *Minerals*, 2021, **11**, 1379.
- 40 J. Rakovský, P. Čermák, O. Musset and P. Veis, *Spectrochim. Acta Part B At. Spectrosc.*, 2014, **101**, 269–287.
- 41 G. S. Senesi, R. S. Harmon and R. R. Hark, *Spectrochim. Acta Part B At. Spectrosc.*, 2021, **175**, 106013.
- 42 B. T. Manard, E. M. Wylie and S. P. Willson, *Appl. Spectrosc.*, 2018, **72**, 1653–1660.
- 43 K. Kjeldahl and R. Bro, *J. Chemom.*, 2010, **24**, 558–564.
- 44 V. Panchuk, I. Yaroshenko, A. Legin, V. Semenov and D. Kirsanov, *Anal. Chim. Acta*, 2018, **1040**, 19–32.
- 45 D. C. C. Montgomery, *Design and Analysis of Experiments*, Wiley, New York, 2017, vol. 2009.
- 46 R. G. Brereton, *Analyst*, 1997, **122**, 1521–1529.
- 47 A. Shulyumova, A. Maltsev and N. Umarova, *X-Ray Spectrom.*, 2018, **47**, 396–404.
- 48 D. Kirsanov, V. Panchuk, M. Agafonova-Moroz, M. Khaydukova, A. Lumpov, V. Semenov and A. Legin, *Analyst*, 2014, **139**, 4303–4309.
- 49 D. Kirsanov, V. Panchuk, A. Goydenko, M. Khaydukova, V. Semenov and A. Legin, *Spectrochim. Acta Part B At. Spectrosc.*, 2015, **113**, 126–131.
- 50 M. D. McKay, R. J. Beckman and W. J. Conover, *Technometrics*, 1979, **21**, 239–245.
- 51 D. M. Steinberg and D. K. J. Lin, *Biometrika*, 2006, **93**, 279–288.
- 52 T. F. Akhmetzhanov, G. V. Pashkova, V. M. Chubarov, T. A. Labutin and A. M. Popov, *J. Anal. At. Spectrom.*, 2021, **36**, 224–232.
- 53 <https://www.oreas.com/>.
- 54 R. G. Brereton, *Applied Chemometrics for Scientists*, John Wiley & Sons, Ltd, Chichester, UK, 2007.
- 55 S. M. Zaytsev, A. M. Popov, N. B. Zorov and T. A. Labutin, *J. Instrum.*, 2014, **9**, P06010.
- 56 S. M. Zaytsev, A. M. Popov and T. A. Labutin, *Spectrochim. Acta Part B At. Spectrosc.*, 2019, **158**, 105632.
- 57 A. M. Popov, T. F. Akhmetzhanov, S. M. Zaytsev, A. N. Drozdova, I. A. Kalko and T. A. Labutin, *J. Anal. At. Spectrom.*, 2022, **37**, DOI: [10.1039/D2JA00127F](https://doi.org/10.1039/D2JA00127F).
- 58 S. Wold, M. Sjöström and L. Eriksson, *Chemom. Intell. Lab. Syst.*, 2001, **58**, 109–130.
- 59 A. L. Pomerantsev, *Chemometrics in Excel*, John Wiley & Sons, Inc., Hoboken, NJ, USA, 2014.
- 60 L. Xu and I. Schechter, *Anal. Chem.*, 1996, **68**, 2392–2400.
- 61 C. M. Andersen and R. Bro, *J. Chemom.*, 2010, **24**, 728–737.
- 62 J. Engel, J. Gerretzen, E. Szymańska, J. J. Jansen, G. Downey, L. Blanchet and L. M. C. Buydens, *Trac. Trends Anal. Chem.*, 2013, **50**, 96–106.
- 63 S. M. Zaytsev, I. N. Krylov, A. M. Popov, N. B. Zorov and T. A. Labutin, *Spectrochim. Acta Part B At. Spectrosc.*, 2018, **140**, 65–72.